

Table 2: Simulation tools for the development of AR, VR, and MR applications

Tool	Platform	Developed in	Features	Open source
Unity3D	Cross-platform	C#, UnityScript	Cross-platform development, Asset store	No
Unreal Engine	Cross-platform	C++, Blueprints	High-fidelity graphics, Blueprint system	No
Vuforia	iOS, Android	C#	Marker-based AR, Cloud recognition	No
ARCore	Android	Java, C++	Motion tracking, Environmental understanding	Yes (GitHub)
ARKit	iOS	Swift, Objective-C	Scene understanding, Face tracking	No
Blender	Cross-platform	Python	3D modelling, Animation	Yes (GitHub)
Spark AR Studio	Windows, macOS	JavaScript	Facebook/Instagram AR effects	No
3ds Max	Windows	MAXScript, Python	3D modelling, animation	No

Table 3: Popular AR, VR, and MR devices available in the market

	Device	Manufacturer	Applications	Resolution in pixels	Price (US\$)
AR	Apple ARKit	Apple	Mobile AR apps, gaming, retail and e-commerce	1920 x 1440	99
	Google Glass Enterprise	Google	Manufacturing, logistics, healthcare, field service	640 x 360	999
VR	Oculus Quest 2	Facebook/Oculus	Gaming, virtual tourism	3664 x 1920 (combined)	299 (64GB)
	Sony PlayStation VR	Sony	Virtual tourism, training, and simulation	1920 x 1080 (combined)	299.99
	HTC Vive Pro 2	HTC	Training and simulation, healthcare	2448 x 2448	799.99
MR	Microsoft HoloLens 2	Microsoft	Industrial design and manufacturing, healthcare	2048 x 1080	3,500
	Magic Leap	Magic Leap	Industrial applications, healthcare	1280 x 960	2,295